

PROBLEM SET 1

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 27/08/2021; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Consider the following examples of “sound” preferences over a space $L(Z)$ of lotteries. For this exercise, you can assume that Z is finite.

Preference for uniformity: The decision maker prefers the less dispersed lottery where dispersion is measured by $\sum_{z \in Z} (p(z) - 1/|Z|)^2$, where $|Z|$ denotes the cardinality, i.e., the number of elements of the prize set Z .

Preference for greatest likelihood: The decision maker prefers p to q if $\max_z p(z)$ is greater than $\max_z q(z)$.

Comparing the most likely prize: The decision maker considers the prize in each lottery that is most likely (breaking ties in some arbitrary way) and compares two lotteries according to a basic preference relation over Z .

For each one of the preference above, check carefully if they satisfy the independence and continuity axioms as defined in lecture 3.

2. A decision-maker expresses the following preference ordering for monetary lotteries (all quantities in thousands of kzt)

$$\begin{aligned} [600] &> [400] > 0.90[600] + 0.10[0] \\ &> 0.20[600] + 0.80[0] > 0.25[400] + 0.75[0] > [0] \end{aligned}$$

Are these preferences consistent with any utility for money? If so, show a utility function that applies. If not, show an axiom that this preference ordering violates.

3. (Based on Markowitz (1959).) One way to construct preference over lotteries with monetary prizes is by evaluating each lottery L on the basis of two numbers: $E[L]$, the expectation of L , and $\text{Var}[L]$, L 's variance. Such a way may or may not be consistent with vNM assumptions. For this exercise, it is fine to consider lotteries over Z in which the set of prizes is finite. Use the version of the axioms in lecture 3.

(a) Show that the function $u(L) = E[L] - (1/4)\text{Var}[L]$ induces a preference relation that is not consistent with the vNM axioms. *Hint: Consider mixtures of each of the lotteries $[1]$ and $0.5[0] \oplus 0.5[4]$ with the lottery $0.5[0] \oplus 0.5[2]$.*

(b) Show that the utility function $u(L) = E[L] - (E[L])^2 - \text{Var}[L]$ is consistent with vNM assumptions.¹ *Hint: It is enough to show that you can find a function $v : Z \rightarrow \mathbb{R}$ such that $u(L) = \sum_z L(z)v(z)$, where $L(z)$ is the probability that lottery L attaches to prize z .*

¹Note that $u(L)$ defined in this way can be written as $u(L) := \mu_L - \mu_L^2 - \sigma_L^2$, where μ_L and σ_L^2 are the mean and variance of the lottery L . If you think of L as the return on an asset portfolio, then this a utility function consistent with vNM assumptions that is defined over the mean return and variance of the portfolio. This is a useful construct for finance; it is widely used in Modern Portfolio Theory.